

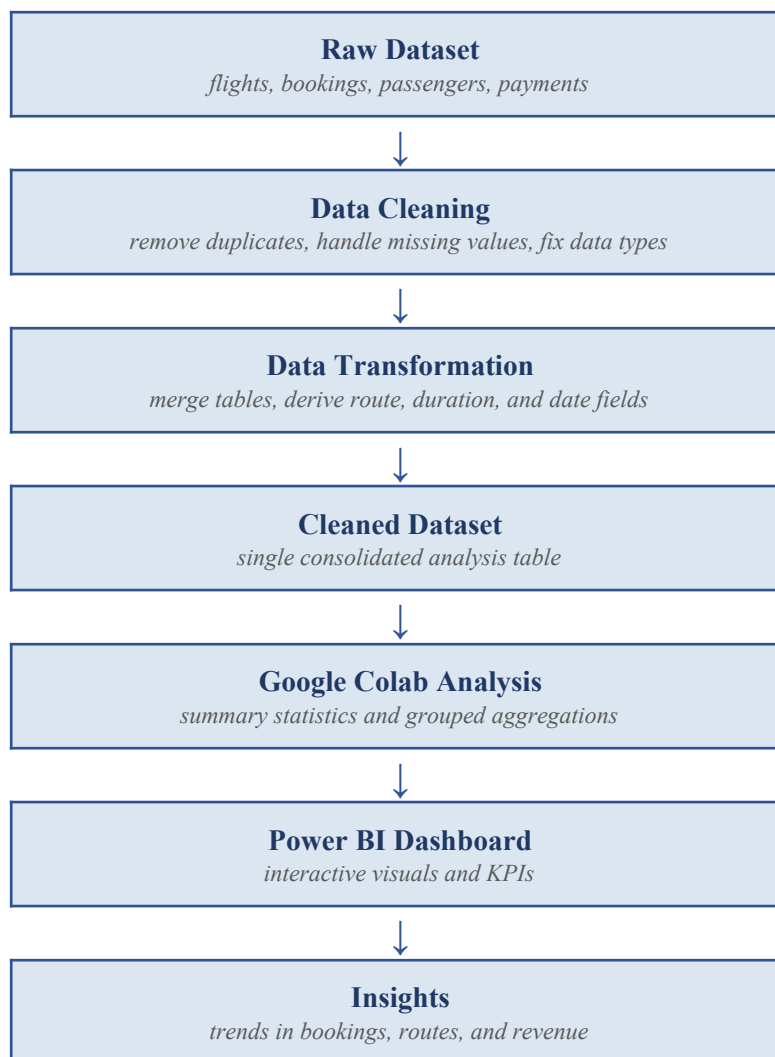
# Airline Booking Analysis

## 1. Project Overview

This project analyzes airline booking data spanning flights, bookings, passengers, and payments. Raw data extracted from an airline operations dataset is cleaned and transformed using Python in Google Colab, then consolidated into a single analytical dataset. The processed data is loaded into Power BI, where an interactive dashboard is built to surface booking trends, route and airline performance, payment behavior, and overall booking status — turning raw operational records into actionable business insights.

## 2. Workflow

The end-to-end workflow moves the data from its raw form through cleaning and transformation to a final, presentation-ready dashboard:



### 3. Data Processing

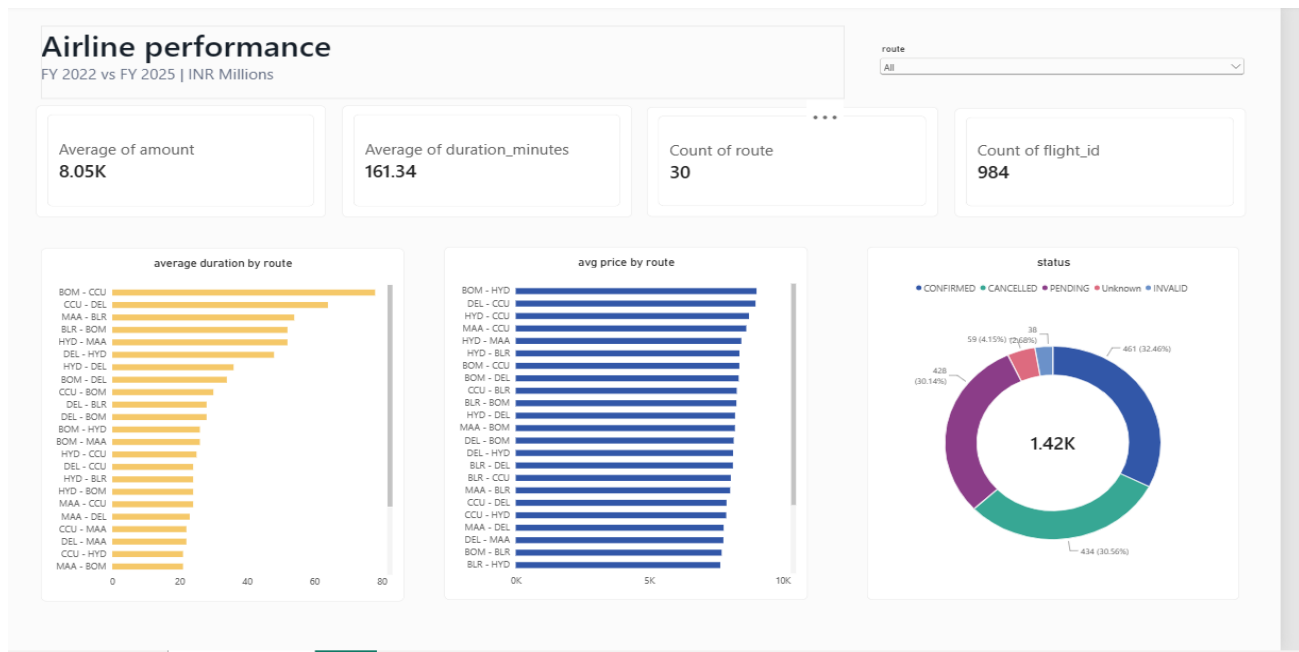
The data was cleaned and processed using **Python (Pandas) in Google Colab**.

- The flights, bookings, passengers, and payments data were loaded from the Excel file.
- Missing values and duplicate records were checked and handled.
- Date, time, and payment columns were converted into the correct format.
- Flight duration was cleaned and converted into minutes.
- The four datasets were combined into one dataset for analysis.
- Personal details such as email, phone number, Aadhaar ID, and passport number were removed for privacy.
- New columns such as route, booking month, and flight date were created.
- Finally, the cleaned dataset was saved as `final_flights_dataset.csv` and used in Power BI

### 4. Tools Used

| Tool         | Purpose                                                                         |
|--------------|---------------------------------------------------------------------------------|
| Python       | Core language used for data cleaning, transformation, and analysis              |
| Google Colab | Cloud notebook environment used to run the Python data-processing pipeline      |
| Pandas       | Library used for reading, cleaning, merging, and reshaping the airline datasets |
| Power BI     | Used to build the interactive dashboard and visualize key metrics               |
| GitHub       | Version control and hosting for the project code and documentation              |

## 5 Screenshot



## 6 Conclusion

This project cleans and analyzes airline booking data — covering flights, bookings, passengers, and payments — using a Python-based pipeline in Google Colab, and presents the results through an interactive Power BI dashboard. The workflow transforms raw, inconsistent operational records into a validated, analysis-ready dataset, and the resulting dashboard makes it possible to explore booking trends, route and airline performance, and payment patterns at a glance, supporting clearer, data-driven decisions.